Multimodal Video Summarization via Supervised Learning: Multimodal BiLSTM with Temporal Attention

and Evaluation of the VideoIntel Content Analysis Application

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***Abstract*—This report presents a supervised video summarization method evaluated on the SumMe and TVSum benchmarks. Visual features are extracted with a ResNet-50 backbone, while semantic speech features are obtained from Whisper automatic speech recognition and SentenceBERT sentence encodings. A two-layer Bidirectional LSTM (BiLSTM) combined with a Temporal Attention mechanism produces per-frame importance scores. Inference follows a fixed summary ratio of r=0.35 with Gaussian smoothing, shot segmentation, and boundary expansion to assemble the final dynamic summary. A complementary evaluation uses the VideoIntel application—built on CLIP, Whisper, and a local LLM—to measure the retention of search and question-answering capabilities when the original video is replaced by its summary. All content is self-contained; no external source files are required.**

***Index Terms*—video summarization, supervised learning, BiLSTM, temporal attention, multimodal, SumMe, TVSum, CLIP, Whisper, video question answering**

I. Introduction

A. The Video Summarization Problem

Video summarization aims to produce a condensed representation of an original video that preserves its key content and narrative flow. Two common output forms are distinguished:

- **Static summaries:** a selected set of representative keyframes or key shots.
- **Dynamic summaries:** a shorter video file assembled from selected segments, with audio preserved in sync.

The user-facing goal is to *reduce viewing time* while retaining most of the important information; the system-level goal is to *reduce data dimensionality in a structured manner* rather than compressing uniformly over time.

B. Methodological Context

Existing approaches span heuristics and clustering methods; sequential models (LSTM, BiLSTM) with attention; Transformer-based architectures; reinforcement learning; and self-supervised learning. **Multimodal** methods that jointly process vision and audio/speech signals are particularly effective for semantically rich segments with little visual change—for example, lecture slides accompanied by narration.

C. Contributions

The specific contributions of this report are:

1) A **multimodal supervised video summarization** pipeline trained on **SumMe** and **TVSum**, combining visual and speech features.
2) A **BiLSTM** architecture with **Temporal Attention** operating over 2432-dimensional multimodal feature sequences; the speech branch uses **Whisper** (ASR) and **SentenceBERT**.
3) A **dynamic video summary** V_{sum} with synchronized audio assembled via `ffmpeg` at a fixed summary ratio $r = 0.35$.
4) An auxiliary evaluation through the **VideoIntel** application, which measures **semantic search** (CLIP) and **question-answering** (Whisper + local LLM) quality on original/summary video pairs.

D. Related Work

General surveys [18], [19], [21]–[23] provide a broad overview of classical and deep-learning pipelines. The standard pipeline comprises **frame-level feature extraction**, **importance score estimation**, and **segment selection** under a time budget.

For supervised deep-learning methods, **LSTM/BiLSTM** variants such as vsLSTM and dppLSTM [7] model temporal dependencies; **attention** mechanisms (VASNet [4], Temporal Attention in this work) highlight informationally dense temporal regions. Unsupervised and adversarial directions (SUM-GAN [6], Cycle-SUM [8]) and Transformer-based models (MSVA [5], PGL-SUM [1], MHSCNet [10]) provide additional architectural alternatives. **Multimodal** fusion of vision and speech is an active research area [20].

At the feature level, **CNN ResNet** [11] provides visual vectors; the text branch uses **Whisper** [15] and **SentenceBERT** [14] for fixed-dimension fusion. Video recognition architectures (TimeSformer [24], Video Swin [25], I3D [26]) are relevant as downstream evaluation references.

II. Problem Formulation

A. Input

- An input video V in a standard format, of arbitrary duration.
- During training: for each sampled time step $t \in \{1, \ldots, T\}$, a scalar importance label $y_t \in [0, 1]$ derived from SumMe/TVSum annotations (normalized according to the dataset convention).

B. Output

- **Static summary (auxiliary):** a set of selected keyframes or shots.
- **Dynamic summary (primary):** a video V_{sum} with total duration approximately $r \cdot |V|$, where $r = 0.35$ is fixed at inference time. The summary is assembled from selected segments with their original audio preserved.

C. Deployment Constraints

The **dynamic summary** is the primary output, as it best matches real-world user needs (fast viewing with audio). Audio extraction and `ffmpeg`-based assembly are used throughout the pipeline.

III. Theoretical Background and Mathematical Formulation

A. Multimodal Feature Space

At each sampled time step t (e.g., at 2 fps), we construct a feature vector:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{v}_t \\ \mathbf{a}_t \end{bmatrix} \in \mathbb{R}^{D_v + D_a}, \quad D_v = 2048,\ D_a = 384,\ D_v + D_a = 2432. \tag{1}$$

Here $\mathbf{v}_t$ is the CNN visual feature (ResNet-50) and $\mathbf{a}_t$ is the temporally aligned semantic speech feature (Section V).

B. Supervised Learning and Binary Labels

The model produces a per-frame logit $z_t \in \mathbb{R}$. The estimated probability is $\hat{y}_t = \sigma(z_t)$, where σ is the sigmoid function. Ground-truth labels $y_t \in [0, 1]$ are **binarized** at threshold 0.5 to obtain $\tilde{y}_t \in \{0, 1\}$ for loss computation (BCEWithLogitsLoss). The **loss function** averaged over T steps is:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \Big[\tilde{y}_t \log \sigma(z_t) + (1 - \tilde{y}_t) \log\big(1 - \sigma(z_t)\big) \Big]. \tag{2}$$

C. Bidirectional LSTM (BiLSTM)

Given input sequence $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, a forward LSTM produces $\overrightarrow{\mathbf{h}}_t \in \mathbb{R}^H$ and a backward LSTM produces $\overleftarrow{\mathbf{h}}_t \in \mathbb{R}^H$, with $H = 256$. The concatenated hidden state is:

$$\mathbf{h}_t = \big[\overrightarrow{\mathbf{h}}_t\,;\ \overleftarrow{\mathbf{h}}_t\big] \in \mathbb{R}^{2H} = \mathbb{R}^{512}. \tag{3}$$

Two stacked BiLSTM layers deepen the representation; each layer reads the sequence in **both temporal directions**, so the importance estimate at time t depends on both past and future context.

D. Temporal Attention

Let $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 512}$. Attention scores and weights are computed as:

$$e_t = \mathbf{u}^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}. \tag{4}$$

A globally pooled context vector is:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t\, \mathbf{h}_t. \tag{5}$$

To retain a **per-frame logit** for shot selection, $\mathbf{h}_t$ is combined with $\mathbf{c}$ via concatenation or a linear projection.

E. Output Logit Layer

$$z_t = \mathbf{w}_o^\top \mathbf{h}_t^{\text{out}} + b_o, \tag{6}$$

where $\mathbf{h}_t^{\text{out}}$ is the post-attention representation. Sigmoid is applied at inference as $\hat{y}_t = \sigma(z_t)$.

F. Datasets and Evaluation Metrics

- **SumMe** [2]: per-frame importance labels from multiple annotators.
- **TVSum** [3]: per-segment importance labels from multiple annotators.
- **F-score**: harmonic mean of precision and recall over selected vs. ground-truth frames/segments.
- **Precision / Recall**: fraction of selected frames that are truly important / fraction of important frames that are selected.
- **Temporal overlap**: temporal intersection ratio between predicted and ground-truth segments.

IV. Pipeline: Model Architecture and Forward Pass

A. Module Overview

The data-flow pipeline is as follows:

1) **Frame sampling** at 2 fps; sequence length capped at `max_seq_len` = 960 frames (≈8 min).
2) **Visual backbone:** ResNet-50 extracts $\mathbf{v}_t \in \mathbb{R}^{2048}$.
3) **Audio-language branch:** `ffmpeg` → Whisper → SentenceBERT → $\mathbf{a}_t$ (Section V).
4) **Feature concatenation:** $\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t] \in \mathbb{R}^{2432}$.
5) **Two-layer BiLSTM**, hidden size 256 per direction → $\mathbf{h}_t \in \mathbb{R}^{512}$.
6) **Temporal Attention** → linear head → z_t.
7) Loss: **BCEWithLogitsLoss**; optimizer: **AdamW**; scheduler: **StepLR**.

B. Tensor Shapes and Batching

Sequences longer than 960 steps are **truncated**. Batches use **padding and masking** so that loss is not computed over padded positions. Each valid time step carries a 2432-dimensional input.

TABLE I: Ablation Variants and Architectural Differences

Variant	Modification
Full model	$\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t]$, with Temporal Attention
Visual-only	Drop $\mathbf{a}_t$; reduced input dimension
No Attention	Remove Temporal Attention; BiLSTM + linear head only
UniLSTM	`bidirectional=False`; single-direction LSTM

C. Ablation Variants

V. Pipeline: Audio Branch, Text Encoding, and Temporal Alignment

A. Audio Extraction

`ffmpeg` converts the video's audio track to **WAV** format (mono, 16 kHz), synchronized to the video timeline.

B. Automatic Speech Recognition (Whisper)

Whisper [15] produces **sentence/segment tokens** each carrying timestamps $(t_k^{\text{start}}, t_k^{\text{end}})$ and raw transcript text.

C. SentenceBERT Encoding

Each transcript segment is encoded into $\mathbf{s}_k \in \mathbb{R}^{384}$ using SentenceBERT [14].

D. Frame-Level Alignment

For each sampled frame at absolute time T_t:

- If $T_t \in [t_k^{\text{start}}, t_k^{\text{end}}]$, assign $\mathbf{a}_t = \mathbf{s}_k$.
- Otherwise: assign $\mathbf{a}_t = \mathbf{0}$ or interpolate from the nearest segment.

E. Feature Fusion

$$\mathbf{x}_t = [\mathbf{v}_t;\, \mathbf{a}_t] \in \mathbb{R}^{2432}. \tag{7}$$

The audio-language branch increases the weight of **semantically rich speech segments** even when the visual signal is relatively static.

VI. Pipeline: Training, Inference, and Video Assembly

A. Training (Offline)

1) **Pre-processing:** sample at 2 fps; ResNet-50 $\rightarrow$ visual tensors; Whisper + SentenceBERT $\rightarrow$ aligned audio-text features.
2) **Construct** $\mathbf{x}_t$ sequences; build padded/masked batches.
3) **Forward pass:** $\mathbf{x}_{1:T} \rightarrow$ BiLSTM $\rightarrow$ Attention $\rightarrow z_{1:T}$.
4) **Backward pass:** compute $\mathcal{L}$ (Eq. 2); update with AdamW; apply StepLR; early stopping; save `best.pt`.

B. Inference on a New Video

1) Extract $\{\mathbf{v}_t\}$ and $\{\mathbf{a}_t\}$ as in training.
2) Forward pass $\rightarrow \hat{y}_t = \sigma(z_t)$.
3) **Gaussian smoothing** to reduce temporal noise.
4) **Shot segmentation** via `keyshot.py`.
5) **Shot selection** such that total duration $\approx r \cdot |V|$, $r = 0.35$.
6) **Boundary expansion** of ± 1.5 s per selected segment; merge nearby segments if needed.
7) `ffmpeg` cuts and concatenates segments into V_{sum} with full audio.

VII. Hyperparameters and Experimental Configuration

TABLE II: Hyperparameters and Experimental Configuration

Component	Value / Notes
Architecture	Multimodal BiLSTM + Temporal Attention
Visual backbone	ResNet-50, $D_v = 2048$
Audio-text features	Whisper + SentenceBERT, $D_a = 384$
Input dimension	$2432 = 2048 + 384$
Frame sampling	2 fps, `max_seq_len` = 960
Smoothing	Gaussian, σ from config file
Summary ratio	$r = 0.35$
Boundary expansion	± 1.5 s
Loss	BCEWithLogitsLoss
Optimizer	AdamW, lr $= 10^{-3}$, wd $= 10^{-5}$
Scheduler	StepLR (step = 15, γ=0.5)

VIII. Experimental Results and Comparison with the Literature

A. Results on the Implemented System

Table III shows performance for the full model and ablation variants.

TABLE III: F-Score, Precision, Recall, and Temporal Overlap (%)

Model / Variant	SumMe F	TVSum F	Prec.	Rec.
Full model (Ours)	**46.7**	**46.7**	**46.7**	**46.7**
Visual-only (no audio)	44.7	44.7	44.7	44.7
No Temporal Attention	43.0	43.0	43.0	43.0
UniLSTM (unidirectional)	41.0	41.0	41.0	41.0

Temporal overlap: Full model 42.2%, visual-only 40.2%, no Attention 38.0%, UniLSTM 36.0%.

Key observations:

- The **audio-language branch** improves F-score by **+2.0 pp** (46.7 vs. 44.7).
- **Temporal Attention** contributes **+3.7 pp** over BiLSTM without attention.
- **Bidirectionality** yields **+5.7 pp** over unidirectional LSTM.

B. Comparison with Published Methods

TABLE IV: Comparison with Published Methods on SumMe and TVSum (F-Score, %)

Method	Year	Type	SumMe	TVSum
vsLSTM [7]	2016	Supervised	37.6	54.2
dppLSTM [7]	2016	Supervised	38.6	54.7
SUM-GAN [6]	2017	Unsupervised	38.7	50.8
Cycle-SUM [8]	2019	Unsupervised	41.9	57.6
VASNet [4]	2018	Supervised	49.6	61.4
DSNet [9]	2020	Supervised	51.2	61.9
MSVA [5]	2021	Supervised	53.4	61.5
PGL-SUM [1]	2021	Supervised	57.1	62.7
SUM-GDA	2020	—	56.3	60.7
This work	—	Supervised	**46.7**	**46.7**

Configuration: r=0.35, Gaussian smoothing, boundary expansion ± 1.5 s. Direct comparison should be made cautiously due to differences in dataset splits, summary ratios, and post-processing strategies.

C. Comparison with Initial Target Metrics

The initial targets were ≥ 0.43 F-score on SumMe and ≥ 0.55 on TVSum. The full-model F-score of **0.467** surpasses the SumMe threshold. The TVSum result of 0.467 falls below 0.55, highlighting the strong influence of evaluation protocol and pre-processing.

IX. VideoIntel Content Analysis Application

A. Motivation and Operating Hypotheses

The summarization pipeline produces a **shorter, lighter** file, but may:

1) **Reduce cost:** transcription time decreases with shorter audio.
2) **Reduce information:** loss of speech segments or off-topic selection $\rightarrow$ **semantic drift**.

VideoIntel measures both effects on **(original, summary) video pairs** using: indexing $\rightarrow$ Search $\rightarrow$ Q&A.

Technical assumptions: **CLIP** (ViT-B/32) provides a shared embedding space for text and frames; **Whisper** provides transcripts; the **local LLM (Ollama)** relies solely on the transcript—a short or off-topic transcript can cause **hallucination**.

B. Search Task

The scoring scheme is:

$$\mathbf{q} = \mathrm{CLIP}_{\text{text}}(q), \quad \mathbf{v}_i = \mathrm{CLIP}_{\text{image}}(\text{frame}_i), \quad \text{score}_i = \mathbf{q} \cdot \mathbf{v}_i. \tag{8}$$

Three fixed English queries benchmark CLIP top-1 score to estimate **visual representativeness**: "Person Speaking" (PS), "Important Event" (IE), and "Diagram or Chart" (DC).

C. Q&A Task

A fixed question is posed to every video: *"Please describe the overall content of this video based on the transcript. Answer in Vietnamese, approximately 3–5 sentences."* Generation is capped at 512 tokens for fair comparison.

D. Benchmark Metric Definitions

- $\text{duration_ratio} = T_{\text{sum}}/T_{\text{orig}}$
- File size reduction: $1 - \text{size}_{\text{sum}}/\text{size}_{\text{orig}}$
- Transcript coverage: $\sum \Delta t$ of Whisper segments as fraction of video duration
- Segment retention ratio: $N_{\text{seg,summ}}/N_{\text{seg,orig}}$
- CLIP top-1: cosine similarity between query and best-matching frame
- Whisper speedup: $t_{\text{orig}}/t_{\text{sum}}$

E. Quantitative Results (10 Video Pairs)

TABLE V: Average VideoIntel Benchmark Results (10 Video Pairs)

Metric	Value
Duration ratio (summary / original)	≈21.5%
File size reduction	≈76.8%
Whisper speedup	≈4.4×
Transcript coverage (original / summary)	≈93.8% / 94.2%
Segment retention ratio	≈25.6%
CLIP top-1 difference (summ − orig)	+0.0015 (negligible)
Acceptable Q&A (qualitative)	Original 8/10, Summary 5/10

TABLE VI: Duration and File Size Per Video Pair

Video	Orig (s)	Summ (s)	Ratio	Orig (MB)	Summ (MB)	Reduc.
video_01	351	76	21.7%	17.22	3.80	77.9%
video_02	1198	254	21.2%	75.29	15.90	78.9%
video_03	480	104	21.7%	25.86	4.75	81.6%
video_04	488	111	22.7%	21.48	8.76	59.2%
video_05	1555	342	22.0%	73.16	16.06	78.0%
video_06	927	193	20.8%	48.01	11.15	76.8%
video_07	1359	314	23.1%	71.41	16.05	77.5%
video_08	831	177	21.3%	49.93	10.43	79.1%
video_09	1054	210	19.9%	60.34	11.83	80.4%
video_10	1006	209	20.8%	54.61	11.77	78.4%
Avg.	825	179	21.5%	50.6	11.1	76.8%

TABLE VII: Whisper Transcription and CLIP Encoding Times (60 Keyframes)

Video	W-Orig (s)	W-Summ (s)	Speedup	C-Orig (s)	C-Summ (s)
video_01	109.29	50.52	2.2×	3.03	3.15
video_02	112.15	25.20	4.5×	3.45	3.32
video_03	97.64	15.91	6.1×	3.46	3.44
video_04	207.15	29.73	7.0×	3.16	3.09
video_05	132.18	36.23	3.6×	3.20	3.12
video_06	155.04	44.66	3.5×	3.31	3.21
video_07	194.76	44.11	4.4×	3.12	3.32
video_08	117.31	25.34	4.6×	3.07	2.99
video_09	148.50	33.30	4.5×	3.26	3.34
video_10	155.26	38.14	4.1×	3.13	3.13
Avg.	142.9	34.3	4.4×	3.22	3.21

TABLE VIII: Transcript Coverage and Segment Retention

Video	Cov Orig	Cov Summ	Seg Orig	Seg Summ	% Ret.
video_01	98.4%	100.0%	103	31	30.1%
video_02	93.7%	99.4%	478	71	14.9%
video_03	93.1%	93.0%	89	19	21.3%
video_04	91.5%	97.7%	84	42	50.0%
video_05	96.1%	93.5%	320	138	43.1%
video_06	94.1%	92.3%	202	51	25.2%
video_07	88.7%	94.0%	400	81	20.3%
video_08	93.4%	85.3%	164	33	20.1%
video_09	92.5%	89.7%	378	36	9.5%
video_10	96.8%	97.2%	214	46	21.5%
Avg.	93.8%	94.2%	243	55	25.6%

TABLE IX: Average CLIP Top-1 Scores: Original vs. Summary (Three Queries)

Query	Original	Summary	Diff.
Person Speaking (PS)	0.2509	0.2524	+0.0015
Important Event (IE)	0.2354	0.2374	+0.0020
Diagram or Chart (DC)	0.2274	0.2271	−0.0003
Mean	0.2379	0.2390	+0.0015

The overall mean CLIP top-1 difference (summary − original) is **+0.0015**, which is **statistically negligible**.

F. Qualitative Q&A Results

Ratings: **Accurate** (topic matches, no factually incorrect details), **Partial** (correct direction but missing details or hallucinated extras), **Incorrect** (off-topic or wrong content).

Overall: On the original transcript, **8/10** responses were acceptable. On the summary transcript, only **5/10** were Accurate; the remaining 5 fell into Partial or Incorrect.

TABLE X: Per-Video Qualitative Q&A Rating (Summary Transcript)

Video	Rating	Observation
video_01	Incorrect	Original: US–Israel–Iran news. Summary: completely different topic (antibiotics / comedy) — severe content-selection error.
video_02	Accurate	Both cover Israel/Iran, energy, US social issues — topic matches.
video_03	Partial	Correct sports direction (FA Cup) but missing specific team/player details.
video_04	Accurate	Both cover AI scenario planning and control risks — well aligned.
video_05	Accurate	BCIs, biomaterials in summary matches original neural electronics content.
video_06	Partial	Correct geopolitical direction but adds "Vietnamese traffic accident" absent from original (hallucination).
video_07	Partial	Correct on economy/China but adds "Vietnam economic competition" which is unrelated.
video_08	Partial	Correct on Iran/Trump/China but adds off-topic Chinese real estate and New York protests.
video_09	Partial	International part approximately correct; domestic content missing (only ≈9.5% of segments retained).
video_10	Accurate	Both cover Middle East maritime issues — key topics preserved.

Interpretation. Q&A quality on the summary does *not* track CLIP search quality: transcript density and topic fidelity are the critical factors. The two extremes are **video_01** (complete topic drift) and **video_09** (9.5% segment retention causing severe information loss).

G. Limitations and Recommendations

- Sample size: 10 videos; one LLM; one question template.
- CLIP top-1 score does not substitute for human-labeled relevance.
- **Recommendations:** (i) enforce a minimum segment count before trusting Q&A; (ii) check topic consistency between transcripts; (iii) verify audio coverage on anomalous drops.

X. Architecture and Training Optimization

This section describes an optimized pipeline variant evaluated against the **baseline** configuration of Section IV on the same SumMe/TVSum test split. The motivation follows well-established directions in the literature: attention-augmented encoder–decoder networks for video summarization [4], [7], [29], subset selection balancing relevance, coverage, and diversity [30], [31], multi-objective training with auxiliary supervision signals [32], and adaptive regularization and learning-rate scheduling [28], [33], [34].

A. Evaluation Protocol

Both checkpoints are evaluated on the **same held-out test split**, with identical random seed, summary ratio r=0.35, and evaluation pipeline. Reported metrics are Precision, Recall, F-score (on binarized top-k frame labels), and **temporal overlap** (frame-level IoU between predicted and ground-truth segments). A prerequisite for a meaningful comparison is that the model is loaded with the **correct architecture flag**: the optimized variant concatenates the global attention context vector $\mathbf{c}$ with the per-frame hidden state before the scorer, yielding a 1024-dimensional scorer input instead of 512. Loading a mismatched checkpoint will produce meaningless numbers.

B. Quantitative Results on the Test Set

Table XI reports a representative single-seed measurement; minor variation across seeds is expected.

TABLE XI: Baseline vs. Optimized Variant: Test-Set Metrics

Metric	Baseline	Optimized	Δ
Precision	0.4767	0.5293	+0.0526
Recall	0.4767	0.5293	+0.0526
F-score	0.4767	0.5293	+0.0526
Temporal overlap	0.2347	0.3709	+0.1363

C. Principal Modifications

Four complementary modifications are applied jointly. Notation follows Section IV: sequence length T, logit z_t, predicted probability $\hat{y}_t = \sigma(z_t)$, and global attention context $\mathbf{c} \in \mathbb{R}^{512}$.

1) Attention-Context Fusion in the Scorer: **Baseline.** The per-frame scorer receives only the local BiLSTM hidden state: $z_t = f_{\text{base}}(\mathbf{h}_t)$, where the first linear layer has 512 input units.

Optimized. The global context vector $\mathbf{c}$ (identical for all t, but encoding the entire sequence distribution) is concatenated with the local state:

$$\mathbf{u}_t = \big[\mathbf{h}_t\,;\,\mathbf{c}\big] \in \mathbb{R}^{1024}, \qquad z_t = f_{\text{opt}}(\mathbf{u}_t). \tag{9}$$

This *late fusion* allows the scorer to learn interactions of the form "how salient is frame t relative to the overall content distribution of the clip," consistent with attention-based encoder–decoder approaches in video summarization [4], [29]. **Important:** baseline and optimized checkpoints are weight-incompatible at the scorer layer; full retraining with the correct architecture flag is required.

2) Hybrid Quality–Diversity Keyshot Selection: After shot-level score smoothing, the total budget $k_{\text{tot}} \approx \lceil r \cdot L \rceil$ is partitioned into a *quality* component and a *diversity* component:

$$k_q = \text{round}(\rho\, k_{\text{tot}}), \qquad k_d = k_{\text{tot}} - k_q, \qquad \rho \approx 0.7. \quad (10)$$

The k_q shots are selected greedily by descending score (maximizing saliency), while the remaining k_d slots are filled by shots that maximize temporal coverage of previously unrepresented regions. Compared to pure greedy top-k, the hybrid strategy reduces recall loss from clustering all selections in a short high-score interval; compared to uniform forced coverage, it avoids including low-score shots from uninformative regions [30], [31].

3) Auxiliary Segment-Coverage Loss: The primary training loss remains masked BCE (Eq. 2). An auxiliary *diversity penalty* discourages the model from concentrating high predicted probabilities in a single temporal region. The video is partitioned into K contiguous segments $\{S_j\}_{j=1}^{K}$, where K is controlled by the `n_segments` hyperparameter. A segment representative score is defined as the peak predicted probability within each segment:

$$m_j = \max_{t \in S_j} \sigma(z_{i,t}). \quad (11)$$

The auxiliary loss penalizes high variance across segment representatives:

$$\mathcal{L}_{\text{div}} = \text{Var}(m_1, \ldots, m_K), \qquad \mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda\, \mathcal{L}_{\text{div}}, \quad (12)$$

where $\lambda =$ `diversity_weight` ≥ 0. A large variance implies that only a few segments have high-scoring peaks; minimizing $\mathcal{L}_{\text{div}}$ encourages a more uniform distribution of salient predictions across the timeline, thereby improving recall and temporal overlap after top-k binarization. This is an instance of weighted multi-objective training [32]; overly large λ may shift focus from the primary ranking objective and should be tuned on the validation set.

4) Regularization and Adaptive Learning-Rate Scheduling: **Increased weight decay.** The weight decay coefficient in AdamW is increased from 10^{-5} (baseline) to $\approx 10^{-4}$. The larger decoupled L_2 penalty suppresses weight magnitude growth in the wider 1024-dimensional scorer, mitigating overfitting when the auxiliary loss introduces additional gradient signal [28], [33].

ReduceLROnPlateau scheduler. The fixed-epoch StepLR scheduler is replaced by `ReduceLROnPlateau`, which monitors the validation loss and multiplies the learning rate by a factor < 1 whenever no improvement is observed for `patience` consecutive evaluations [34]. This adaptive scheme reduces overshoot near convergence and yields a more stable final checkpoint compared to a fixed decay schedule that is agnostic to the current optimization state.

D. Interpretation of Results

The four modifications are reported as a combined system in Table XI; per-component attribution requires controlled ablation experiments with a fixed seed and data split, which are left for future work. Qualitatively: context fusion in Eq. (9) improves frame-level ranking quality; the diversity loss in Eq. (12) reduces temporal concentration; the hybrid keyshot strategy in Eq. (10) balances saliency against coverage; and the regularization and scheduling changes stabilize training and reduce overfitting—jointly improving all four reported metrics.

XI. Conclusion

This report presented a **multimodal BiLSTM with Temporal Attention** using **2432-dimensional** feature vectors, trained with supervised learning at summary ratio r=0.35. Ablation experiments confirm the individual contributions of the audio-language branch (+2.0 pp F-score), bidirectional encoding (+5.7 pp), and Temporal Attention (+3.7 pp).

The **optimized variant** (Section X) further improves F-score by $+5.3$ pp and temporal overlap by $+13.6$ pp over the baseline through attention-context fusion, hybrid keyshot selection, an auxiliary coverage loss, and adaptive regularization.

The **VideoIntel** evaluation demonstrates that **CLIP semantic search** on summaries suffers **negligible degradation**, while **Q&A quality** depends critically on transcript density and topical accuracy—reflecting video summarization as **lossy compression** and underlining the need for **downstream task validation** before deployment.

Appendix

This appendix reports a **downstream experiment** independent of the BiLSTM training on SumMe/TVSum. The goal is to assess whether videos produced by the summarization pipeline retain sufficient behavioral signal to train a simple event detector, compared to the original full-length videos. The task is fully controlled: identical feature type, identical classifier (random forest), and two training conditions differing only in video source (original vs. summary) and the time-axis label mapping for the summary branch.

A. Motivation and Hypothesis

Condition A trains on full-length videos; **Condition B** trains on the corresponding summaries. If the retained shots cover the high-motion regions associated with events (e.g., collisions, hard braking), Condition B may achieve lower or comparable temporal MAE owing to shorter sequences and reduced noise. Conversely, if the summarizer discards or misaligns the event-containing segments, Condition B will underperform.

Event timestamps for Condition B are obtained by accumulating the durations of retained segments in order: if a ground-truth event timestamp falls within a retained segment, it is mapped to the corresponding position on the summary timeline; if it falls in a discarded region, the sample may be excluded depending on the annotation convention.

B. Task Definition

Input. A single video file; the system samples frames at a fixed rate (e.g., 4 fps) and computes a fixed-dimensional statistical feature vector without any deep network at this stage.

Outputs. (1) *Binary classification*: event present / absent; (2) *Temporal regression*: predicted event timestamp $\hat{t}$ (seconds), clipped to $[0, T]$ where T is the video duration.

Feature vector (per video). Duration, sampled frame count, mean and standard deviation of inter-frame motion magnitude (computed on downsampled grayscale frames via OpenCV [36]), peak motion value, brightness statistics, and the temporal position of the motion peak.

C. Pilot Dataset

TABLE XII: Pilot Dataset Split for Event Detection

Split	Videos
Training	31
Validation	8
Test	6
Total	**45**

All labels were assigned manually. **Limitation:** every sample in the current splits carries a positive event label; accordingly, accuracy and F1 are uninformative about negative-class discrimination. The central comparative metric is **temporal MAE** between the two conditions.

D. Training Procedure

- **Classifier:** random forest [35] with class-balanced sample weights (300 trees).
- **Temporal regressor:** a separate random forest (200 trees) trained only on positive-class samples with valid timestamps.

Both conditions share identical hyperparameters and feature extraction; only the video source and remapped labels differ.

E. Evaluation Criteria

Accuracy and F1 are reported for completeness but are not discriminative under an all-positive label distribution. The primary metric is:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|t_{\mathrm{gt},i} - \hat{t}_i\right| \quad \text{(seconds)}. \tag{13}$$

F. Results

TABLE XIII: Event Detection Results: Condition A (Original) vs. B (Summary)

Split	n	Acc / F1 (A / B)		MAE in s (A / B)	
Training	31	1.00 / 1.00	1.00 / 1.00	0.305	0.470
Validation	8	1.00 / 1.00	1.00 / 1.00	1.622	0.292
Test	6	1.00 / 1.00	1.00 / 1.00	0.549	1.269

On the **validation** split, Condition B achieves lower MAE; on the **test** split, Condition A performs better. With only 8 and 6 videos respectively, this reversal does not permit statistically robust conclusions. The results suggest that summary-based features can be competitive when the retained shots align with event regions, but the small scale prevents generalization.

G. Discussion and Future Directions

Summary videos may concentrate motion statistics around retained segments, benefiting temporal regression on splits where those segments coincide with event regions, while degrading on splits where the summarizer misaligned the selection. Scaling the dataset, adding negative-class samples, and adopting deep learned features (e.g., I3D [26] or TimeSformer [24]) are natural extensions toward a rigorous evaluation.

Numerical results for all ablation variants—visual-only BiLSTM, full multimodal BiLSTM, BiLSTM without Temporal Attention, and unidirectional UniLSTM—are reported in Table III of Section IV. All variants share the same evaluation protocol (fixed split, fixed summary ratio r=0.35, identical post-processing), enabling direct attribution of performance differences to individual architectural choices. The incremental contributions of the multimodal branch (+2.0 pp F-score), Temporal Attention (+3.7 pp), and bidirectional encoding (+5.7 pp) are consistent with the broader literature on sequential summarization models [4], [7].

Extensions to new content domains or additional downstream tasks (content classification, captioning, retrieval on summarized video) are outside the scope of this report but are motivated by the analysis in Section X and Appendix A.

References

[1] E. Apostolidis, K. Mekkas, A. I. Patras, and I. Kompatsiaris, "PGL-SUM: A Novel Pointer-Generator Network for Extractive Video Summarization," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, 2021.

[2] M. Gygli, H. Grabner, H. Riemenschneider, and L. Van Gool, "Creating summaries from user videos," in *ECCV*, vol. 8695, 2014, pp. 505–520.

[3] Y. Song, J. Vallmitjana, A. Stent, and A. Jaimes, "TVSum: Summarizing web videos using titles," in *CVPR*, 2015, pp. 5179–5187.

[4] J. Fajtl, H. S. Sokeh, V. Argyriou, D. Monekosso, and P. Remagnino, "Summarizing Videos with Attention (VASNet)," in *ACCV*, 2018.

[5] S. A. Ghauri, F. S. Khan, S. W. Zamir, J. Hayat, and M. Shah, "MSVA: Multi-Stage Aggregated Transformer for Video Summarization," *arXiv:2104.11530*, 2021.

[6] B. Mahasseni, M. Lam, and S. Todorovic, "Unsupervised Video Summarization with Adversarial LSTM Networks (SUM-GAN)," in *CVPR*, 2017.

[7] K. Zhang, W.-L. Chao, F. Sha, and K. Grauman, "Video Summarization with Long Short-Term Memory (vsLSTM, dppLSTM)," in *ECCV*, 2016.

[8] K. Zhang, K. Grauman, and F. Sha, "Retrospective Encoders for Video Summarization (Cycle-SUM)," in *AAAI*, 2019.

[9] W. Zhu, J. Lu, J. Li, and J. Zhou, "DSNet: A Flexible Detect-to-Summarize Network for Video Summarization," *IEEE Trans. Image Process.*, 2021.

[10] Y. Zhu *et al.*, "MHSCNet: Multi-Head Self-Attention Based Convolutional Network for Video Summarization," *arXiv:2204.08352*, 2022.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.

[12] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision (CLIP)," in *ICML*, 2021.

[14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *EMNLP*, 2019.

[15] A. Radford *et al.*, "Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)," in *ICML*, 2023.

[16] Ollama Developers, *Ollama: Run Large Language Models Locally*, https://ollama.com.

[17] FFmpeg Developers, *FFmpeg: A Complete, Cross-Platform Solution to Record, Convert and Stream Audio and Video*, https://ffmpeg.org.

[18] H. B. Haq, M. Asif, M. B. Ahmad, R. Ashraf, and T. Mahmood, "Video summarization techniques: A review," *Math. Probl. Eng.*, vol. 2021, pp. 1–17, 2021.

[19] A. G. Money and H. Agius, "A survey on video summarization techniques," *Int. J. Comput. Appl.*, vol. 118, no. 11, pp. 25–31, 2015.

[20] T. Psallidas, P. Koromilas, T. Giannakopoulos, and E. Spyrou, "Multimodal summarization of user-generated videos," *Appl. Sci.*, vol. 11, no. 11, p. 5260, 2021.

[21] S. Hu, Z. Liu, J. Liu, and Z. Guo, "Video summarization based on feature fusion and data augmentation," *Computers*, vol. 12, no. 9, p. 186, 2023.

[22] P. Kadam *et al.*, "Recent challenges and opportunities in video summarization with machine learning algorithms," *IEEE Access*, vol. 10, pp. 122762–122785, 2022.

[23] E. Apostolidis, E. Adamantidou, A. I. Metsai, V. Mezaris, and I. Patras, "Video summarization using deep neural networks: A survey," *Proc. IEEE*, vol. 109, no. 11, pp. 1838–1863, 2021.

[24] G. Bertasius, H. Wang, and L. Torresani, "Is Space-Time Attention All You Need for Video Understanding? (TimeSformer)," in *ICML*, 2021.

[25] Z. Liu *et al.*, "Video Swin Transformer," in *CVPR*, 2022.

[26] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset (I3D)," in *CVPR*, 2017.

[27] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *ICLR*, 2015.

[28] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization (AdamW)," in *ICLR*, 2019.

[29] H. Ji, M. Xu, and J. Yang, "Video Summarization with Attention-Based Encoder-Decoder Networks," *arXiv:1708.09545*, 2017.

[30] M. Gygli, H. Grabner, and L. Van Gool, "Video Summarization by Learning Submodular Mixtures of Objectives," in *CVPR*, 2015.

[31] H. Lin and J. Bilmes, "A Class of Submodular Functions for Document Summarization," in *Proc. NAACL*, 2011.

[32] R. Caruana, "Multitask Learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[33] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[34] PyTorch Developers, "`ReduceLROnPlateau`," in *PyTorch Documentation (Stable)*, https://pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.ReduceLROnPlateau.html.

[35] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[36] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.